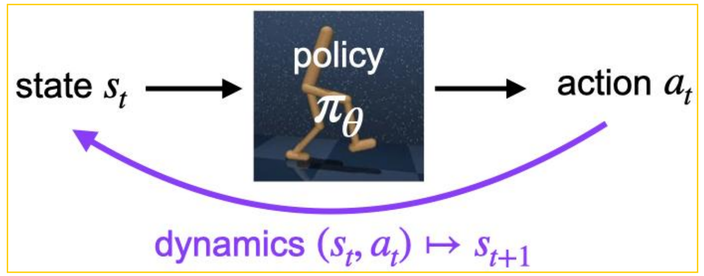
Butterfly Effects of SGD Noise:
Error Amplification in Behavior Cloning and Autoregression

Adam Block♦, Dylan Foster♠, Akshay Krishnamurthy♠, Max Simchowitz♦, Cyril Zhang♠

♦MIT ♠Microsoft Research NYC

Behavior cloning & feedback loops



BC: fit θ to a collection of demonstrations from expert policy $\pi^\star$, minimizing 1-step loss

(e.g.) $\ell_{\mathrm{BC}}(\theta) = \mathbb{E}_{\text{expert's } (s_t, a_t)} \left[\|\pi_\theta(s_t) - \pi^\star(s_t)\|^2 \right]$

Usual view: BC reduces offline imitation learning to supervised learning

$\ell_{\mathrm{BC}}(\theta) \le \varepsilon$ (train on 1-step loss) → $J_H(\pi_\theta) \ge J_H(\pi^\star) - C \cdot \varepsilon$ (evaluate multi-step rollout rewards)

Q1: *When BC works, how big is C in practice?*

Q2: *When BC fails, must we improve the data? Or can we improve upon standard training?*

Special case: pretraining vs. generation in LLMs

s_t : sequence of t tokens

a_t : append a single token

π_θ : autoregressive language model

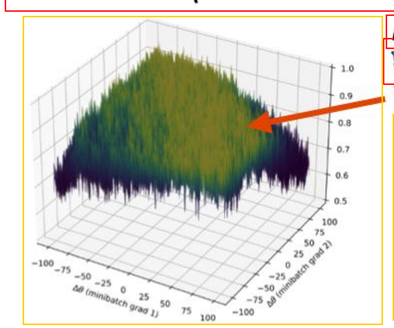
Gradient Variance Amplification

A1: *Long-horizon rollouts can be extremely sensitive to small SGD fluctuations.*

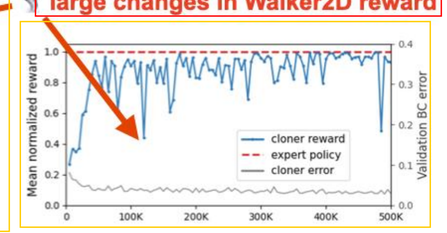
GVA: an empirical pathology in BC with NNs

- *small changes in θ due to minibatch noise*
- *...causing small changes in 1-step $\ell_{\mathrm{BC}}(\theta)$...*
- *...and **chaotic** long-rollout rewards $J_H(\pi_\theta)$*

Evidence: *(across locomotion tasks & archs)*



In stochastic gradient directions:
- small changes in ℓ_{BC}
- large changes in Walker2D reward



fractal reward landscape **oscillations during training**

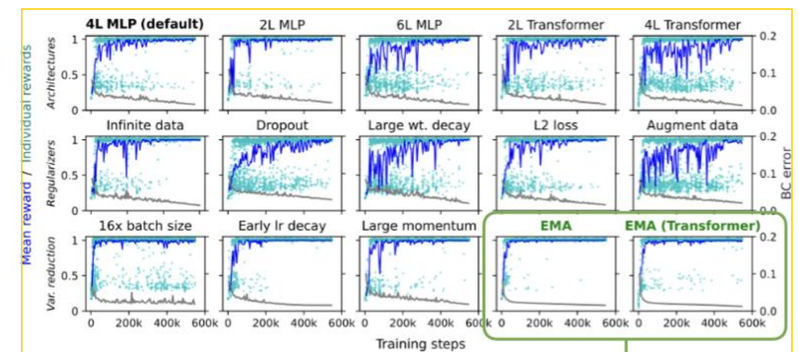
(in the paper: accompanying quantitative studies)

Theory: vignettes in toy models

- marginally stable linear dynamics, linear $\pi, \pi^\star$
- "CliffLQR": halfspace reward → GVA
- SDE limit: EMA & LR schedules both work

Algorithmic mitigations for GVA

A2: *Variance reduction techniques mitigate the oscillations; EMA is highly effective.*

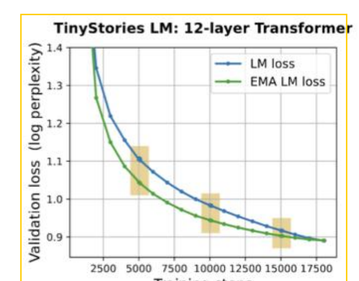


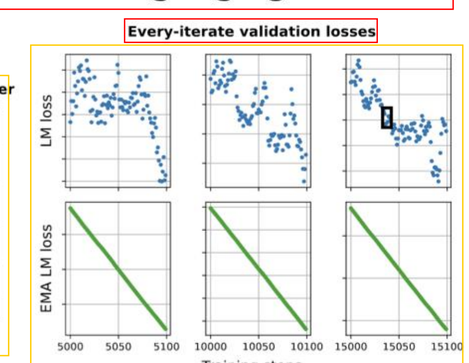
Exponential Moving Average

$\theta_{t+1} \leftarrow \theta_t - \eta_t \widetilde{\nabla}_\theta \ell_{\mathrm{BC}}(\theta)$ (or Adam, etc.)

$\bar{\theta}_{t+1} \leftarrow (1 - \beta_t)\bar{\theta}_t + \beta_t \cdot \theta_t$, use $\bar{\theta}_t$ for inference

Bonus: GVA & EMA in language generation





EMA improves & stabilizes LM iterates before learning rate cooldown